

Exponents & Radicals

Student learning objectives for this unit

SUPPORTING SKILLS LEGEND

I — Introduce

D — Deepen

A — Apply

R — Reinforce

☐ 1. I can **simplify** expressions containing negative exponents.

RELATED SKILLS

☐ I Rewrite a negative exponent using a reciprocal.☐ A Apply the product rule for exponents.☐ A Apply the quotient rule for exponents.

☐ 2. I can **rewrite** expressions between radical and rational-exponent forms.

RELATED SKILLS

☐ I Convert between radical and rational-exponent notation.☐ I Interpret numerator and denominator roles in a rational exponent.☐ I Read radical and rational-exponent notation and identify the index, radicand, base, numerator, and denominator.

☐ 3. I can **simplify** radical expressions.

RELATED SKILLS

☐ I Extract perfect-power factors from a radical.☐ A Express a whole number as a product of prime factors.

☐ 4. I can **apply** exponent rules to expressions with integer and rational exponents.

RELATED SKILLS

☐ I Apply the zero-exponent rule.☐ D Apply the power rule for exponents.☐ D Apply the product rule for exponents.☐ D Apply the quotient rule for exponents.☐ A Interpret numerator and denominator roles in a rational exponent.☐ A Rewrite a negative exponent using a reciprocal.

☐ 5. I can **solve** a basic equation involving a radical or rational exponent and check for extraneous solutions.

RELATED SKILLS

☐ I Check a candidate solution for extraneous results.☐ I Isolate a radical expression before solving.☐ I Require an even-root radicand to be nonnegative when determining domain.☐ A Use inverse operations to maintain an equivalent equation.